



HOW TO USE THIS SET

20 questions, mixed difficulty, 7 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.

EASY

Which of the following liver enzymes is considered the most liver-specific marker of hepatocyte injury?

- A. Amylase
- B. GGT
- C. AST
- D. ALP
- E. ALT

2.

EASY

Total bilirubin is best described as:

- A. A clotting factor made by the liver
- B. A byproduct of red blood cell breakdown and the pigment responsible for jaundice
- C. A protein synthesised only by the kidney
- D. An enzyme released only from bone
- E. A marker of pancreatic function

3.

EASY

Which protein, synthesised by the liver, is the most abundant plasma protein and typically falls in chronic liver disease?

- A. Albumin
- B. C-reactive protein
- C. Fibrinogen only
- D. Globulin
- E. Haemoglobin

4.

EASY

PT/INR primarily evaluates which coagulation pathway?

- A. The extrinsic pathway (factors II, V, VII, X)
- B. The complement pathway
- C. The intrinsic pathway
- D. Platelet aggregation only
- E. The fibrinolytic pathway only

5.

MEDIUM

CASE

A 58-year-old man presents with jaundice, pruritus, pale stools and dark urine. Labs: AST 98 IU/L, ALT 110 IU/L, ALP 850 IU/L, total bilirubin 13 mg/dL (mainly conjugated), albumin 4.1 g/dL, INR 1.3. Ultrasound shows a gallstone in the common bile duct.

This biochemical pattern is best classified as:

- A. Obstructive (cholestatic) pattern
- B. Pure haemolytic pattern
- C. Isolated coagulopathy without liver disease
- D. Normal liver function
- E. Hepatocellular pattern

6.

MEDIUM

CASE

A 30-year-old woman with a history of paracetamol overdose develops confusion and jaundice. Labs: AST 2300 IU/L, ALT 2550 IU/L, ALP 135 IU/L, total bilirubin 8.5 mg/dL, albumin 3.7 g/dL, INR 4.

This pattern, with massively raised transaminases and a markedly prolonged INR plus encephalopathy, is most consistent with:

- A. Gallstone obstruction
- B. Chronic stable cirrhosis
- C. Acute fulminant hepatic failure
- D. Isolated bone disease
- E. Pre-renal azotaemia

7.

MEDIUM

An AST:ALT ratio greater than 1 (AST higher than ALT) is classically associated with which condition?

- A. Viral hepatitis
- B. Bone disease
- C. Gallstone obstruction alone
- D. Alcoholic liver disease
- E. Vitamin K deficiency

8.

MEDIUM

ALP is raised in liver, bone, bile duct and placental disease because ALP is:

- A. Non-specific, being produced by liver, bone, bile ducts and placenta
- B. Produced only by hepatocytes
- C. Only elevated in pregnancy
- D. Highly liver-specific
- E. A marker exclusively of chronic alcohol use

9.

MEDIUM

GGT is most useful clinically because, when raised together with ALP, it helps confirm that the source of the raised ALP is:

- A. Liver or biliary tract rather than bone
- B. Bone rather than liver or biliary tract
- C. Kidney only
- D. Skeletal muscle
- E. Placenta specifically

10.

MEDIUM

A prolonged PT/INR in a patient with liver disease most directly reflects:

- A. Reduced synthesis of vitamin K-dependent clotting factors by the diseased liver
- B. Increased platelet count
- C. Increased albumin synthesis
- D. Reduced bilirubin conjugation
- E. Increased GGT activity

11.

MEDIUM

Unconjugated (indirect) bilirubin differs from conjugated (direct) bilirubin in that unconjugated bilirubin is:

- A. Only produced after cholecystectomy
- B. Undetectable in healthy individuals
- C. Water-soluble and excreted directly in urine
- D. Bound to albumin and not yet processed by the liver
- E. The dominant form in obstructive jaundice

12.

MEDIUM CASE

A patient on long-term warfarin therapy is found to have a prolonged PT/INR on routine bloodwork, with normal liver enzymes and normal albumin.

What is the most likely explanation for the prolonged INR in this patient?

- A. Isolated hyperbilirubinaemia
- B. Chronic alcohol-related AST elevation
- C. Obstructive biliary disease
- D. Underlying hepatocellular disease
- E. Warfarin's inhibition of vitamin K-dependent clotting factor synthesis, independent of liver damage

13.

MEDIUM

Which combination of results would you expect to see rising together in chronic liver disease due to increased immunoglobulin production?

- A. GGT only
- B. Gamma-globulins (part of the globulin fraction)
- C. Albumin and PT
- D. Unconjugated bilirubin only
- E. ALT only

14.

MEDIUM

Liver function tests are used for all of the following EXCEPT:

- A. Diagnosing and monitoring liver disease severity
- B. Evaluating hepatotoxicity from medications
- C. Directly visualising bile duct anatomy
- D. Screening for alcohol- or toxin-related liver damage
- E. Investigating unexplained jaundice or fatigue

15.

HARD CASE

A 45-year-old man with long-standing alcohol use presents with fatigue. Labs show AST 140 IU/L, ALT 65 IU/L, ALP 110 IU/L, GGT 210 IU/L, albumin 3.2 g/dL, INR 1.1.

Which feature of this panel most strongly supports an alcohol-related aetiology rather than viral hepatitis?

- A. AST exceeds ALT and GGT is disproportionately elevated
- B. ALP is markedly elevated
- C. Albumin is elevated
- D. INR is markedly prolonged
- E. Bilirubin is entirely unconjugated

16.

HARD CASE

A patient has a low albumin, mildly prolonged INR, and normal bilirubin, ALT and AST, in the setting of nephrotic syndrome.

What is the most likely explanation for the low albumin in this scenario, given the normal transaminases?

- A. Vitamin K deficiency
- B. Chronic hepatocellular damage reducing synthesis
- C. Obstructive biliary disease
- D. Renal loss of albumin (nephrotic syndrome) rather than impaired hepatic synthesis
- E. Acute fulminant hepatic failure

17.

HARD

A patient has ALP markedly elevated with normal GGT. Which source of the raised ALP is most likely, and why does the normal GGT matter?

- A. Liver, because GGT is not typically used to localise ALP elevation
- B. Bile duct obstruction, because GGT is never useful in cholestasis
- C. Placenta, because GGT is always elevated in pregnancy
- D. Bone, because GGT is largely confined to liver/biliary/kidney tissue and stays normal when ALP rises from a non-hepatobiliary source
- E. Pancreas, because GGT tracks pancreatic enzyme release

18.

HARD

CASE

Two patients both have total bilirubin of 12 mg/dL. Patient A's bilirubin is 90% unconjugated; Patient B's is 90% conjugated.

Which statement correctly distinguishes the likely underlying processes?

- A. Patient A has obstructive disease; Patient B has haemolysis
- B. Neither pattern has clinical significance
- C. Patient A's pattern suggests increased haemolysis or impaired hepatic uptake/conjugation; Patient B's pattern suggests impaired excretion, as in biliary obstruction or hepatocellular disease affecting excretion
- D. Both patterns suggest identical pathology
- E. Conjugated bilirubin is always higher in haemolytic anaemia

19.

HARD

In a patient with cirrhosis, which combination of laboratory findings reflects synthetic liver failure most specifically, as opposed to simple hepatocellular inflammation?

- A. Isolated GGT elevation from alcohol use
- B. Isolated ALT elevation
- C. Elevated AST alone without other abnormalities
- D. Low albumin and prolonged INR, reflecting reduced synthesis of albumin and clotting factors
- E. Normal albumin with high ALP

20.

HARD

CASE

A patient's LFTs show AST 3200 IU/L, ALT 3450 IU/L, ALP 120 IU/L, bilirubin 9 mg/dL, albumin 3.5 g/dL, INR 1.2, with no encephalopathy and rapid clinical improvement over the following week.

Compared with the fulminant hepatic failure pattern described in the lecture (INR 4, encephalopathy), what does the near-normal INR and absence of encephalopathy here suggest about the liver's synthetic function?

- A. Synthetic function is severely and irreversibly impaired despite the transaminase rise
- B. Synthetic function (clotting factor production) has been preserved despite marked hepatocyte injury, distinguishing this from fulminant failure
- C. The bilirubin level rules out any hepatocellular injury
- D. ALP must be falsely low due to lab error
- E. The pattern indicates chronic cirrhosis rather than acute injury

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	E	ALT
2	B	A byproduct of red blood cell breakdown and the pigment responsible for jaundice
3	A	Albumin
4	A	The extrinsic pathway (factors II, V, VII, X)
5	A	Obstructive (cholestatic) pattern
6	C	Acute fulminant hepatic failure
7	D	Alcoholic liver disease
8	A	Non-specific, being produced by liver, bone, bile ducts and placenta
9	A	Liver or biliary tract rather than bone
10	A	Reduced synthesis of vitamin K-dependent clotting factors by the diseased liver
11	D	Bound to albumin and not yet processed by the liver
12	E	Warfarin's inhibition of vitamin K-dependent clotting factor synthesis, independent of liver damage
13	B	Gamma-globulins (part of the globulin fraction)
14	C	Directly visualising bile duct anatomy
15	A	AST exceeds ALT and GGT is disproportionately elevated
16	D	Renal loss of albumin (nephrotic syndrome) rather than impaired hepatic synthesis
17	D	Bone, because GGT is largely confined to liver/biliary/kidney tissue and stays normal when ALP rises from a non-hepatobiliary source
18	C	Patient A's pattern suggests increased haemolysis or impaired hepatic uptake/conjugation; Patient B's pattern suggests impaired excretion, as in biliary obstruction or hepatocellular disease affecting excretion
19	D	Low albumin and prolonged INR, reflecting reduced synthesis of albumin and clotting factors
20	B	Synthetic function (clotting factor production) has been preserved despite marked hepatocyte injury, distinguishing this from fulminant failure

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.